

Resources and Development

Social Science — Geography • Chapter 1 • Chapter Summary & Study Guide for Daksh

What does a tank of fuel, a wooden table, your meal of dal-rice and the air you breathe have in common? They all started as **resources** — things in the environment that we use to satisfy our needs. But not everything in nature is a resource: it must be technologically accessible, economically viable and culturally acceptable. This chapter looks at how we classify resources, why over-use has created problems, and why we must plan and conserve them — especially the most fundamental of all, **land and soil**.

Classifying resources

Geographers classify resources in several useful ways:

- **By origin:** Biotic (forests, fishery, cattle) vs Abiotic (rocks, water, air).
- **By exhaustibility:** Renewable (sun, wind, water) vs Non-renewable (coal, petroleum, minerals — formed over millions of years).
- **By ownership:** Individual (private farms), Community (village pasture, parks), National (within country boundaries — minerals, oceans within 200 km), International (oceans beyond territorial limits).
- **By status of development:** Potential (Rajasthan's solar/wind potential), Developed (already in use), Stock (known, but no technology to use yet), Reserves (part of stock that can be used in future, e.g. river-water for hydel).

Development of resources — and its problems

Resource development has powered every modern economy. But indiscriminate use has caused:

- **Depletion** of resources for short-term gain.
- **Concentration** of resources in few hands → divided society into haves and have-nots.
- **Ecological crises** — global warming, ozone depletion, pollution, land degradation.

The **Rio Earth Summit (1992)** at Brazil and **Agenda 21** declared **sustainable development** the way forward — meeting present needs without compromising the ability of future generations to meet theirs.

Resource planning in India

India has been doing resource planning since its First Five-Year Plan (1951). It involves three steps:

1. Identification & inventory of resources across regions
(surveying, mapping, qualitative & quantitative estimation, measurement)
2. Evolving a planning structure
(appropriate technology, skill set, institutional setup)
3. Matching resource development with national development plans

Why does India need this? Because resources are unevenly distributed — Rajasthan is rich in solar/wind but lacks water; Jharkhand and Odisha have coal and iron ore; Ladakh has rich culture but few mineral resources. Planning ensures balanced regional development.

Land — a fundamental resource

India has a wide variety of land — mountains (~30%), plateaus (~27%), plains (~43%). Land is used for forests, agriculture, pastures, settlements, roads and industry. **Land use pattern** changes over time as population, technology and economic priorities change. Areas of concern: agricultural land share has not grown as much as needed, and forest cover (~22%) is below the 33% target set by the National Forest Policy.

Soil — types and conservation

- **Alluvial soil** — most fertile; deposited by rivers in the Northern Plains and deltas; rich in potash, lime; ideal for wheat, rice, sugarcane. *Khadar* = newer; *Bangar* = older.

- **Black soil (Regur)** — Deccan trap region (Maharashtra, Gujarat, MP); from weathered basaltic lava; rich in lime, calcium, magnesium; great for cotton.
- **Red & Yellow soil** — found over crystalline igneous rocks in low-rainfall regions of the Deccan; reddish due to iron oxides.
- **Laterite soil** — humid tropical areas with heavy rainfall (Karnataka, Kerala, MP); leached of nutrients; suitable for tea, coffee, cashew with manuring.
- **Arid soil** — Rajasthan, Gujarat; sandy, low humus and moisture; with irrigation supports millets.
- **Forest soil** — hilly regions; loamy and silty in valleys; coarser on slopes.

Soil erosion and conservation

Soil erosion = removal of topsoil by wind/water/human activity faster than nature can replace it. Caused by deforestation, overgrazing, mining, faulty agricultural practices. **Conservation methods:** **contour ploughing** (along contour lines on slopes), **terrace farming** (steps on hillsides), **strip cropping** (alternating crop strips), **shelter belts** (rows of trees in deserts), **afforestation**, controlled grazing, regulated mining.

★ MUST-REMEMBER POINTS

- Resource = environmental input that meets a need (technologically accessible, economically viable, culturally acceptable).
- Renewable (sun, wind, water) vs Non-renewable (coal, petroleum, minerals).
- Sustainable Development → Rio Earth Summit 1992 + Agenda 21.
- Resource planning steps: identify, build planning structure, match with national plans.
- Soil types: alluvial (rivers, fertile); black/regur (Deccan, cotton); red-yellow (low rainfall); laterite (heavy rain, leached); arid (deserts); forest (hills).
- Khadar = new alluvial, Bangar = old alluvial.
- Soil conservation: contour ploughing, terrace farming, shelter belts, afforestation.

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